

MGT 430: BIOTECHNOLOGY INDUSTRY, STRUCTURE & STRATEGY

Summer 2026

MBA FW

Room: OTRSN 1E106

Class Hours: Flex Hybrid Weekend Saturday, 1:00 – 5:00

PROFESSOR: Robert Bain

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OFFICE HOURS: By appointment

COURSE DESCRIPTION AND GOALS

Biotechnology started as a science, referring to the use of living cells as factories to produce protein through manipulation of genes. However, today biotechnology refers to an industry, with the top companies in the sector exceeding some of the major pharmaceutical companies in market capitalization. No longer are biotechnology companies constrained to using recombinant DNA technology alone, as the moniker is now assigned to any small company engaged in life sciences-related research directed toward developing a commercial product using a wide range of scientific approaches.

These products are not necessarily limited to therapeutics but may include diagnostic or medical device products. Belonging to the sector also implies a culture—small, nimble, visionary yet practical, cash-constrained, and willing to take significant risks. While some of these characteristics are idealized, it is clear that although development-stage companies rely on the same scientific, analytical, and managerial talent found in large pharmaceutical and device companies, the context is fundamentally different. Success requires additional skills, adaptability, and non-traditional systems.

The foundation of this course builds on the core curriculum you have completed during your first year at Rady. You will have an advantage relative to many industry professionals in your ability to apply structured thinking and analysis.

However, the industry presents unique challenges. Long development cycles can reduce urgency in decision-making, and the ability to operate at a loss for extended periods can distort incentives. In some cases, promotion may take precedence over rigorous analysis, and short-term share price considerations may overshadow long-term value creation.

This course is designed to avoid these traps. You will be expected to think across functions, adapt to changing conditions, and operate with limited margin for error. Your personal decision-making style will be tested in a highly visible and dynamic environment.

COURSE FORMAT

The goal of this course is not to make you experts on industry facts and figures. Instead, the objective is to teach you to identify issues, rigorously analyze them, and develop actionable solutions from the perspective of a senior manager in a biotechnology enterprise.

The course focuses on running a biotechnology company rather than starting one. It combines discussion and lecture, anchored around real-world case studies.

You will be exposed to all major functional areas, including discovery, development, manufacturing, regulation, marketing, and sales. Broader topics such as strategy, financing, licensing, intellectual property, bioethics, reimbursement, and organizational behavior will also be covered.

The course includes three in-person Saturday sessions and online modules. In-person sessions will feature guest lecturers and student-led case discussions. Sessions run from 1:00 pm to 5:00 pm with scheduled breaks.

Given the breadth of content and limited time, some prioritization is required. While medical devices and diagnostics will be discussed, the primary focus will be on pharmaceutical applications due to their complexity.

Students will frequently take on leadership roles in discussions. Additional formats may include team exercises and guest speakers.

Throughout the course, you are expected to evaluate situations based on the information available at the time and make decisions from the perspective of a senior executive operating under real-world constraints.

HOMEWORK AND GRADING

This course is designed with high expectations. Feedback from prior classes has been incorporated to continually improve both depth and breadth.

Class participation and online assignments account for 35% of your grade. Students are expected to thoroughly prepare assigned cases, ideally working with their teams.

The course includes students with varying levels of experience, contributing diverse perspectives. Use this to your advantage.

In addition to informal participation, students will receive formal feedback through:

- Leading case discussions
- Company analysis assignments

Students will receive feedback multiple times throughout the course, either verbally or in writing. Please proactively seek feedback if needed.

Grades will reflect both absolute performance and relative standing within the class. Final grades are not calculated as an average, but as an overall assessment of performance and effort.

Historically, grades cluster around A-/B+, with A/A+ reserved for top performers.

You will be evaluated not on your ability to summarize information, but on the quality of your thinking, your judgment, and your ability to make and support decisions.

FORMAL ASSIGNMENTS – CASE DISCUSSIONS

Each student will participate in a case team responsible for leading at least one case discussion.

Case teams are expected to:

- Provide a concise summary of the case
- Identify key issues or opportunities
- Evaluate strategic options
- Deliver a recommendation supported by analysis and rationale

This is not a descriptive exercise. Students are expected to demonstrate analytical thinking and decision-making as if they are senior executives or advisors to the board.

During discussions, questions may be directed to case team members to deepen analysis and initiate broader engagement. The remainder of the class will participate as board members or investors.

All students are expected to actively participate in all case discussions, regardless of whether they are part of the presenting team.

TEAM STRUCTURE AND CASE ASSIGNMENTS

Students will be assigned to teams of 2 to 5 depending on class size. Each team will be assigned one or more cases as outlined below:

- Moderna (Technology Discovery)
- Theranos (Drug and Technology Development)
- Antegren (Manufacturing)
- ZappRx (Finance)
- Molecular Insight Pharmaceuticals – MIP (IP)
- Tangion (HEOR and Reimbursement)
- Proteus (Licensing)
- Flow Tech (Marketing and Sales)
- Gardasil (Pricing – All Teams)

Team assignments, case allocations, and the Team's live case presentation schedule will be posted on Rady Canvas once the course roster has been finalized.

PREPARATION AND EXPECTATIONS

Students are expected to read and prepare all assigned cases in advance of each session.

For those presenting during the first session, an outline on how to structure the case discussion will be provided in Rady Canvas. This outline is intended to guide your analysis and clarify expectations.

Students are encouraged to reach out in advance with any questions.

As outlined in the course bibliography, there is a significant amount of assigned reading in addition to the cases. Required readings are identified in black and recommended readings in blue. Additional materials may be provided throughout the course.

Students are expected to determine which information is most relevant to addressing the key issues in each case. You should approach each case as a senior executive responsible for making decisions, not as a passive reviewer of information.

This means:

- Identifying the critical issues
- Evaluating trade-offs across potential solutions
- Developing a clear and actionable plan

Students are encouraged to go beyond the case materials and incorporate external sources, including company websites, public filings, and discussions with individuals who have relevant experience.

You will be evaluated based on problem-solving and decision-making abilities rather than memorization of content. Consistent with this philosophy, there are no exams in this course.

CASE EVALUATION FRAMEWORK

Students must evaluate each case within the context of the time period in which it was written. You should not rely on information, outcomes, or solutions that became available after the case timeframe.

This is not an exercise in hindsight. Rather, you are expected to assess the situation based on what was known or reasonably knowable at that time.

You should approach each case as if you are making decisions in real time under uncertainty. The objective is to determine what you would do if you were in their position, given the information, constraints, and uncertainties that existed at that point.

These assignments are designed to reinforce analytical thinking, independent judgment, and the ability to connect strategic decisions to real-world outcomes, including financial impact.

ATTENDANCE

Due to the limited number of sessions, attendance is expected at all classes. If an unavoidable conflict arises, please notify me in advance.

COMPANY ASSIGNMENT (65% OF FINAL GRADE)

The remaining 65% of your grade will be based on a Company Assignment paper submitted in three installments over the duration of the quarter, with the final installment due the evening of the last session.

The objective of this assignment is to apply course concepts through the in-depth analysis of a life sciences company. Each student will select a company with which they have no more than a passing familiarity. This is an individual assignment, and each student must select a unique company. A sign-up sheet will be available on Rady Canvas to confirm selections.

Students are expected to utilize both publicly available information and primary research through interviews. At least two interviews must be conducted with individuals connected to the company or its ecosystem. These may include company executives, competitors, industry analysts, academic experts, or other relevant stakeholders.

Interviews must be based solely on publicly available information and general industry perspectives. Students should not seek or request any confidential, proprietary, or non-public information. Under no circumstances should a student request or agree to sign a non-disclosure agreement (NDA) as part of this assignment. This is not an exercise in obtaining confidential information, but in applying analytical thinking to publicly available data and informed perspectives.

This is not a descriptive report, but an analytical assessment supported by evidence. Students are expected to form independent perspectives that may differ from company messaging or public narratives.

Students should leverage high-quality sources where available, including SEC filings, earnings calls, investor presentations, scientific publications, and other credible materials. Interview insights must be integrated into the analysis and not presented as a separate section.

INSTALLMENT 1 – COMPANY PROFILE

Provide a comprehensive overview of the company, including:

- Products and/or services
- Target markets and customers
- Competitive positioning
- Financial and operational performance

The goal is to establish a clear understanding of the company's business model, strategy, and current state.

INSTALLMENT 2 – KEY ISSUES OR OPPORTUNITIES

Identify three distinct strategic issues or opportunities facing the company.

These must be:

- Specific and company-relevant
- Actionable and decision-oriented
- Grounded in the company's unique context, including its technology, leadership, market dynamics, and investor landscape

Broad or generic industry trends are not sufficient unless clearly tied to company-specific impact.

Students must:

- Clearly define each issue or opportunity
- Prioritize the three items
- Provide a rationale for the ranking

After prioritization, select one issue or opportunity for deeper analysis. This does not need to be the top-ranked item, but the choice should be justified.

INSTALLMENT 3 – FINAL ANALYSIS AND RECOMMENDATION

Evaluate how the company is currently addressing the selected issue or opportunity.

Based on your research and interviews:

- Assess the effectiveness of the company's approach
- Identify gaps, risks, or assumptions in their strategy
- Provide your own perspective on how the issue or opportunity should be addressed
- Assess the potential financial impact of both the company's approach and your recommended alternative, including implications on revenue, cost structure, profitability, valuation, or market positioning where applicable

Students should clearly state whether they agree or disagree with the company's approach and support their position with data, analysis, and interview insights.

Your recommendation should reflect how a senior executive or investor would evaluate and act on this issue, including the expected financial implications of that decision.

Additional assignment details will be provided on Rady Canvas.

COURSE SCHEDULE EXAMPLE (Assignment due dates are in blue)

Day	Module	Lectures	Case Studies	Case Study Groups	Company Assignment
27-Jun	Module 0	Class Assignments Overview			
4-Jul	Module 1	Strategy and Structure	Moderna	Group 1	
11-Jul	Module 2	Drug/Tech Development	Theranos	Group 2	
18-Jul	Module 3	Manufacturing	Antegren	Group 3	
18-Jul		In-person Saturday Guest Lecturers	Case Study Presentations	Groups 1, 2 & 3	1st Company Assignment Due
25-Jul	Module 4	Financing	ZappRx	Group 4	
1-Aug	Module 5	IP	MIP	Group 5	
8-Aug	Module 6	Licensing	Proteus	Group 6	
15-Aug		In-person Saturday Guest Lecturers	Case Study Presentations	Groups 3, 4, & 5	2nd Company Assignment Due
12-Aug	Module 7	HEOR	Tangion	Group 7	
29-Aug	Module 8	Sales/Marketing	Flow Tech	Group 8	
5-Sep	Module 9	Pricing	Gardasil	All Groups	
12-Sep		In-person Saturday Guest Lecturers	Case Study Presentations Gardasil Case Study	Groups 7 & 8 All Groups	Final Company Assignment Due

LOGISTICS

In-person course meeting dates are July 18th, August 15th, and September 12th. The Saturday sessions will begin at 1:00 am and conclude at 5:00 pm. Breaks will be provided throughout the day.

Please direct all questions via e-mail at rbain@UCSD.edu or through Rady Canvas. This syllabus may change, and additional details about each session will be added as needed in advance. Please check Rady Canvas regularly for up-to-date information. I will also use Rady Canvas to communicate with you. My alternative contact information is listed on the first page of the syllabus. I will be available for office hours by appointment at one of the faculty offices at Rady or scheduled remotely. Please arrange in advance.

ACADEMIC INTEGRITY AND USE OF GENERATIVE AI

Integrity of scholarship is essential to the academic community. As members of the Rady School of Management students and faculty are expected to uphold the highest ethical standards and protect the integrity of the University's intellectual work. All academic work submitted in this course must reflect the student's own independent effort and analysis unless otherwise authorized.

The intent of this course is to develop strategic thinking analytical reasoning decision making under uncertainty and the ability to assess real world biotechnology business situations using public information research and interviews. Generative AI tools can be used transparently for research or editing, but must not replace original analysis, critical thinking, assignment creation, or interviews.

The following guidelines apply to the acceptable and unacceptable use of generative AI in this course:

- Students are required to disclose the use of generative AI tools in the preparation of assignments.
- Generative AI may be used for limited support functions such as grammar formatting summarization of public information and preliminary background research.
- Students may not submit AI generated analysis conclusions recommendations interview content or interpretations as their own original work.
- Core assignment components including strategic analysis conclusions recommendations interviews and interpretation of business considerations must reflect the student's independent thinking and judgment.
- Students remain fully responsible for the accuracy originality and integrity of all submitted work.

The complete UCSD Policy on Integrity of Scholarship can be viewed at:

<http://www.senate.ucsd.edu/manual/appendices/app2.htm#AP14>

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.**

Please visit the [OSD website](#) for further information or contact the Office for Students with Disabilities at (858) 534-4382 or OSD@ucsd.edu.

SATISFACTORY ACADEMIC PROGRESS (SAP)

Satisfactory Academic Progress (SAP) refers to the academic standards students must maintain to remain eligible for federal, state, and institutional financial aid. **If you are receiving financial aid, please ensure you review the [SAP requirements and the appeals process](#).**

BOOKS: No Text is required for this class.

Optional texts:

Pisano, Gary. [Science Business](#). Harvard Business School Press (2006).

This book, published by an HBS professor, is the closest to a “textbook” for the course. While there will be no formal reading assignments from the book, and it is somewhat dated, I recommend reading it.

Mellon, Jim and Chalabi, Al. [Cracking the Code](#). John Wiley & Sons, Ltd (2012)

This book, looks at the industry written not as scientists but as investors, investigating over 100 companies and is packed with wise and practical advice. (Highly recommend)

Bazell, Robert. [Her-2: The Making of Herceptin, a Revolutionary Treatment for Breast Cancer](#). Random House (1998).

The story of the first drug based on the shift to “pharmaco-genomics”, providing insight on how Genentech became a market leader in cancer by being the scientific innovator.

Robbins-Roth, Cynthia. [From Alchemy to IPO: The Business of Biotechnology](#). Perseus Books Group (May 15, 2000).

This book is more specific to the small company biotechnology setting and would be useful background information.

Werth, Barry. [The Billion Dollar Molecule: One Company's Quest for the Perfect Drug](#). Simon & Schuster (1995).

This is probably the best general interest book ever written about biotech. The book follows a company, Vertex Pharmaceuticals, from its inception through its “growing pains”, and ultimately to its success. Like a good thriller, it has it all – good guys versus bad guys, heart-stopping action, and pathos. Enjoy!

You will receive digital copies of all required and recommended readings, in your course reader. Hard copies of other relevant readings that are not available on-line will be kept in the library, reserved for your use. Any updates to the syllabus will be posted on Rady Canvas.

WEEKLY SCHEDULE GUIDELINE:

Online, SATURDAY JUNE 27

- **MODULE 0: CLASS INTRODUCTION AND ASSIGNMENTS OVERVIEW**

Online, SATURDAY JULY 4

- **MODULE 1: STRATEGY AND BUSINESS MODELS (Video Lecture)**

During this first session, (after an orientation to the logistics of the course), we will discuss the biotechnology industry's structure and strategies, in the form of examination of various business models – their rationale and sustainability. The session will also stress the economics of the industry as a whole, as well as of individual functions therein.

CASE:

Marco Iansiti, Karim R. Lakhani, Hannah Mayer, Kerry Herman “Moderna (A)” Harvard Business School #621032-PDF-ENG, Sep 15, 2020(Revised: Jul 13, 2021): 1-20.

Explores how a biotech company modernizes and expedites the drug/vaccine discovery and development process from years (multiple) to months.

OTHER ARTICLES:

[Pisano, Gary P., Stephanie Oestreich, Clarissa Ceruti. “The Life Sciences Revolution: A Technical Primer.” Harvard Business School, #9-602-118 \(rev. August 6, 2002\): 1-15. \(A background piece on the science underlying biotechnology.\)](#)

Online, SATURDAY JULY 11

- **MODULE 2: DRUG/TECHNOLOGY DEVELOPMENT (Video Lecture)**

During this part of the session, we will discuss the clinical development process. To those trained as scientists, clinical development seems so simple, yet it is the most complex, risky, and expensive part of the business. In this area, the company must make tradeoffs between rigor and practicality, in a process that is ultimately responsible for whether a company will succeed or fail. In development, outstanding execution is necessary but not sufficient for success. Strategic and tactical planning (in advance) is what separates the winners from the losers.

CASE:

Fuller, Joseph B. and Masko, John. “Theranos: The Unicorn that Wasn’t.” Harvard Business Publishing, #319068 (Revised: September 6, 2019)

An overview on the development and collapse of a startup company built on vapor technology and fraud. The interesting part about this case, are all the players and their involvement

OTHER ARTICLES:

[Aspinall, M. G. and Hamermesh, R. G. “Realizing the Promise of Personalized Medicine” Harvard Business School, R0710F, Oct 01, 2007](#)

[Guttmacher, A.E., and Collins, F.S. “Genomic Medicine – A Primer” *The New England Journal of Medicine*, November 7, 2002, Vol 347 No. 19. Pg 1512-1520](#)

Online, SATURDAY JULY 18

▪ **MODULE 3: MANUFACTURING (Video Lecture)**

Unlike in other industries, where manufacturing is all about lowering cost of goods, manufacturing plays a more strategic role in the biotechnology industry. It is the key element used by regulators to define and approve classic biotechnology products (biologics) and is critical to establishing barriers to entry of others.

CASE:

Margolis, J., DeLong, T. and Heymann, T. “Antegren: A Beacon of Hope”, Harvard Business School, #9-408-025 (November 14, 2007) 1-17

Biogen Idec addresses critical manufacturing decisions in the face of a product launch which has the potential to significantly affect the company’s fortunes and patients’ quality of life.

Wouters, McKee, and Luyten “Estimated Research and Development Investment Needed to Bring a New Medicine to Market, 2009-2018” JAMA. 2020;323(9):844-853, Corrected on September 20, 2022.

IN-PERSON 1:00-5:00, SATURDAY JULY 18

GUEST LECTURE ROUNDTABLE / SPEED DATING

CASE STUDY PRESENTATION: Module 1

Marco Iansiti, Karim R. Lakhani, Hannah Mayer, Kerry Herman “Moderna (A)” Harvard Business School #621032-PDF-ENG, Sep 15, 2020(Revised: Jul 13, 2021): 1-20.

Explores how a biotech company modernizes and expedites the drug/vaccine discovery and development process from years (multiple) to months.

CASE STUDY PRESENTATION: Module 2

Fuller, Joseph B. and Masko, John. “Theranos: The Unicorn that Wasn’t.” Harvard Business Publishing, #319068 (Revised: September 6, 2019)

An overview on the development and collapse of a startup company built on vapor technology and fraud. The interesting part about this case, are all the players and their involvement

CASE STUDY PRESENTATION: Module 3

Margolis, J., DeLong, T. and Heymann, T. “Antegren: A Beacon of Hope”, Harvard Business School, #9-408-025 (November 14, 2007) 1-17

Biogen Idec addresses critical manufacturing decisions in the face of a product launch which has the potential to significantly affect the company’s fortunes and patients’ quality of life.

Online, SATURDAY JULY 25

- **MODULE 4: FINANCING (Video Lecture)**

It takes years for most biotechnology companies to become profitable. It takes hundreds of millions of dollars of investment to get a company to that point. Yet even though money may be available to a company, knowing when and how to collect it is a key decision for management.

CASE: Busgang, Jeffrey and Hull, Olivia. "ZappRx." Harvard Business Publishing, #818001 January 18, 2018 (Revised: January 15, 2019)

This case typifies some of the key challenges in financing a new biotech company, not least of which is valuation and the trade-offs between various options.

OTHER ARTICLES:

Adams, C.P. and Brantner, V. V. "Estimating the Cost of New Drug Development: Is It Really \$802 Million?" *Health Affairs*, 25, no. 2 (2006): 420-428

Hall, Sam & Alastair J.J. Wood, "Financial Growing Pains of a Biotech," *The Scientist*, August 2008, Vol. 22, Issue 3, pp. 30-36.

ONLINE, SATURDAY AUGUST 1

- **MODULE 5: INTELLECTUAL PROPERTY (Video Lecture)**

In this session, you will be introduced to both legal and business elements of intellectual property (IP), and how company operations and strategy need to take IP into account. We will use the case of Prometheus to examine the key issues of publish vs. patent; who owns intellectual property rights; and when should legal action be pursued to protect a patent.

CASE:

Hemermesh, Richard G. "Molecular Insight Pharmaceuticals, Inc." Harvard Business Publishing, #805067 January 12, 2005 (Revised: March 1, 2006)

IP is a key asset for any biotechnology company and this case highlights issues of intellectual property protection in the context of a health care venture.

OTHER ARTICLES:

SUPREME COURT OF THE UNITED STATES: OCTOBER TERM 2011, Syllabus, MAYO COLLABORATIVE SERVICES, DBA MAYO MEDICAL LABORATORIES, ET AL. v. PROMETHEUS LABORATORIES, INC. (Assigned reading)

Webber, Philip M. "Protecting Your Inventions: The Patent System." *Nature Reviews Drug Discovery*, vol. 2, no. 10 (October 2003): 823-830.

Shuchman, Miriam. "Delaying Generic Competition – Corporate Payoffs and the Future of Plavix." *The New England Journal of Medicine*, Vol. 355, No. 13 (September 28, 2006): 1297-1300.

ONLINE, SATURDAY AUGUST 8

▪ **MODULE 6: HEALTH ECONOMICS OUTCOMES RESEARCH (Video Lecture)**

In the past decade one of the most significant shifts in the market for healthcare technologies has been the shift in decision-making power away from the healthcare provider to the payer and the patient. The ability to clearly demonstrate the value of new healthcare technologies is a key strategic issue for all biotechnology companies, irrespective of size, sector or stage of development.

CASE:

Ofek, E. and Ross Ribatt, P. "Tengion: Bringing Regenerative Medicine to Life" Harvard Business School, #9-510-031 (rev. April 14, 2010): 1-29.

This is a case about Tengion, a company with an interesting challenge, trying to find a balance between their business strategy, available financing, and their mission to alleviate patient suffering.

OTHER ARTICLES:

[Lindgardt, Zhenya, Martin Reeves & Judith Wallenstein. "Waking the Giant: Business Model Innovation in the Drug Industry." *In Vivo*, vol. 26, no. 6 \(June 2008\): 54-59.](#)

[Cutler, David M. "The Demise of the Blockbuster?" *New England Journal of Medicine*, Vol. 356, no. 13 \(March 29, 2007\): 1292-1293.](#)

IN-PERSON, SATURDAY AUGUST 15

GUEST LECTURE ROUNDTABLE / SPEED DATING

CASE STUDY PRESENTATION: Module 4

Bussgang, Jeffrey and Hull, Olivia. "ZappRx." Harvard Business Publishing, #818001 January 18, 2018 (Revised: January 15, 2019)

This case typifies some of the key challenges in financing a new biotech company, not least of which is valuation and the trade-offs between various options.

CASE STUDY PRESENTATION: Module 5

Hemermesh, Richard G. "Molecular Insight Pharmaceuticals, Inc." Harvard Business Publishing, #805067 January 12, 2005 (Revised: March 1, 2006)

IP is a key asset for any biotechnology company and this case highlights issues of intellectual property protection in the context of a health care venture.

CASE STUDY PRESENTATION: Module 6

Ofek, E. and Ross Ribatt, P. "Tengion: Bringing Regenerative Medicine to Life" Harvard Business School, #9-510-031 (rev. April 14, 2010): 1-29.

This is a case about Tengion, a company with an interesting challenge, trying to find a balance between their business strategy, available financing, and their mission to alleviate patient suffering.

ONLINE, SATURDAY AUGUST 22

▪ **MODULE 7: LICENSING (Video Lecture)**

For most early-stage biotechnology companies, the likelihood that they will see the development of their product through to commercialization as a stand-alone company is slight. On the other side of the coin, large pharmaceutical, device and biotechnology companies are more than ever reliant on innovation from outside their own laboratories to fuel their need for new products. Licensing, therefore, is of critical importance to any and all players in the biotechnology marketplace.

CASE:

Hamermesh, R.G., Barley, L. and Graham, G., "Proteus Biomedical: Making Pigs Fly", Harvard Business School, #9-809-051 (April 2, 2010) 1-22

This case looks at the need to "make a deal" from the perspective of a small biotech company with more good ideas than money.

OTHER ARTICLES:

Aghazadeh, Behzad, Catherine Arnold, Tong Zhang, Charley Beever, and Jeff Boschwitz. "Making Licensing Pay." *In Vivo: The Business and Medicine Report*, vol. 23, no. 5 (May 2005): 43-47.

Arnold, Katie, Anthony Coia, Scott Saywell, Ty Smith, Scott Minich, Alicia Loffler. "Value Drivers in Licensing Deals." *Nature Biotechnology*, vol. 20 (November 2002): 1085 – 1089.

ONLINE, SATURDAY AUGUST 29

▪ **MODULE 8: MARKETING & SALES (Video Lecture)**

The decisions surrounding the launch of a new pharmaceutical product are of significant proportions. While the product has been slowly marching through the minefield of development towards marketing approval the decision to launch is viewed by many as a foregone conclusion, yet there are many strategic options available to the manufacturer regarding the launch, including a decision not to launch. With the hugely competitive market for healthcare technologies, the investment in building a sales and marketing capability for the product is substantial, and the decisions surrounding the sales and marketing resources have enormous impact on the commercial success of the product.

CASE:

Herzlinger, R.E., "I've Got Rhythm: Selling Cardiac Rhythm Management Devices", Harvard Business School, #9-304-012 (August 14, 2006) 1-30

This case will give us the chance to discuss key questions regarding marketing and sales strategy in a highly competitive market for medical devices.

OTHER ARTICLES:

Antonuccio, David O., David D. Burns, and William G. Danton, "Antidepressants: A Triumph of Marketing Over Science?" *Prevention & Treatment*, July 15, 2002, Vol. 5, No. 25.

ONLINE, SATURDAY September 5

▪ **MODULE 9: PRICING**

The prior session on Health Economics and Market Access provides a natural segue into a discussion on pricing of new healthcare technologies, through the analysis of the economic value of the product. Pricing has the single biggest impact on the return on investment from a new healthcare technology and getting the launch price right is of critical importance, as the ability to increase the price of a pharmaceutical product after launch is significantly constrained.

Additionally, the issue of pricing of new healthcare technologies in the current environment of significant resource constraints brings forth much discussion about the ethics of drug pricing. While there will be no time to devote specifically to the important issue of bioethics, this topic will likely find its way into many of our class discussions, in particular the session on pricing. There are no specific articles or texts in the reading list on this topic but those who are interested in this topic may make a request for suggestions.

CASE:

Calkins, T. "Merck: Pricing Gardasil" Harvard Business School, KEL400 (2009) 1-10

OTHER ARTICLES:

Farrington, George, "How to Price Cancer Drugs." *In Vivo* (July/August 2007): 47-53.

Berenson, Alex, "Cancer Drugs Offer Hope, But at Huge Expense," *The New York Times*, July 12, 2005, p. A.1.

IN-PERSON, SATURDAY SEPTEMBER 12

GUEST LECTURE ROUNDTABLE / SPEED DATING

CASE: Module 7

Hamermesh, R.G., Barley, L. and Graham, G., "Proteus Biomedical: Making Pigs Fly", Harvard Business School, #9-809-051 (April 2, 2010) 1-22

This case looks at the need to "make a deal" from the perspective of a small biotech company with more good ideas than money.

CASE: Module 8

Herzlinger, R.E., "I've Got Rhythm: Selling Cardiac Rhythm Management Devices", Harvard Business School, #9-304-012 (August 14, 2006) 1-30

This case will give us the chance to discuss key questions regarding marketing and sales strategy in a highly competitive market for medical devices.

FINAL CASE (Pricing): Module 9

Calkins, T. "Merck: Pricing Gardasil" Harvard Business School, KEL400 (2009) 1-10

GENERAL INFORMATION RESOURCES^{1,2,3}

Journals:

- Nature Biotechnology; <http://www.nature.com/nbt> (especially the *BioEntrepreneur* column).
- Nature Reviews Drug Discovery; <http://www.nature.com/nrd>.
- New England Journal of Medicine; <http://content.nejm.org> (the most prestigious general medical journal, with frequent articles on policy issues).

Trade Publications:

- BioCentury; <http://www.biocentury.com> (daily news, and weekly feature articles with strategic bent).
- BioWorld; <http://www.bioworld.com> (daily news).
- Medical Technology Stock Letter. <https://bioinvest.com> (Biotechnology stock-picking advice by one of the best in the business)
- Red Nucleus; <https://rednucleus.com/medcomms/>

Data Sources:

- Analyst Reports from Investment Banks (only available to account holders)
- Biospace; <http://www.biospace.com>.
- Ernst & Young Annual Reports on Life Sciences; <http://www.ey.com> (free to clients).
- IQVIA; <https://www.iqvia.com> (the best source for data on markets).
- Medscape; <http://www.medscape.com> (general reviews and on-line continuing education on various medical topics, organized by specialty).

Books:

- Biotech Industry: A Global, Economic, and Financing Overview. Bryan Bergeron and Paul Chan. Wiley (January 23, 2004).
- Biotech Investing: Every Investor's Guide. Jim McCamant. Perseus Publishing (May 2002).
- Building Biotechnology: Starting, Managing, and Understanding Biotechnology Companies. Yali Friedman. thinkBiotech LLC (2004).
- The Business of Healthcare Innovation. Ed. Lawton Robert Burns. Cambridge University Press (2005).
- The Coming Biotech Age: The Business Of Bio-Materials. Richard W. Oliver. McGraw-Hill (2000).
- The Golden Helix: Inside Biotech Ventures. Arthur Kornberg. University Science Books (1995).
- Kellogg on Biotechnology. Alicia Loffler. Northwestern University Press (2005).
- Multinational Pharmaceutical Companies: Principles and Practices. Bert Spilker. Lippincott Williams & Wilkins; 2nd edition (January 15, 1994). *Everything you always wanted to know about the pharmaceutical industry*.
- Redefining Healthcare. Michael E. Porter and Elizabeth Olmsted Teisberg. Harvard Business School Press (2006).
- Startup to IPO: How to Build and Finance a Technology Company. Donald H Macadam. Xlibris (February 2, 2004).
- US Biotechnology VC Directory. BioAbility, LLC. BioWorld Publishing Group (2003).
- Your Money or Your Life: Strong Medicine for America's Health Care System. David M. Cutler. Oxford University Press (January 7, 2005).

Government / Industry Sites:

- BIO – Biotechnology Industry Organization; <https://www.bio.org> (national biotechnology industry trade organization).

- BIOCOM; <http://www.biocom.org> (California regional biotechnology industry trade organization).
- FDA; <http://www.fda.gov>
- KFF; <https://www.kff.org> (The independent source for health policy research, polling, and news)
- PhRMA.org; <http://www.phrma.org> (pharmaceutical industry trade organization).
- PubMed (Medline); <http://www.ncbi.nlm.nih.gov> (to search within the scientific/medical literature).
- US Patent Office; <http://www.uspto.gov>.

1. Free resources (of which there are not many) are noted in bold; 2. I will try to make sure that as many of the journals and books listed above as possible are available (in some form) at the library; 3. ALL ARE OPTIONAL!

▪ **MODULE 3: HEALTH TECHNOLOGY REGULATION:**

The biotechnology industry is very heavily regulated and understanding how best to navigate the maze of regulation through the various stages of development to commercialization is a key strategic capability. A knowledgeable and experienced regulatory affairs team can shave significant time, effort and investment off the development process and thereby significantly improve a company's return on investment.

CASES:

Diermeier, D. "Reintroduce Thalidomide? (A)" Harvard Business School, KEL058 (2006) 1-16

Diermeier, D. "Reintroduce Thalidomide? (B)" Harvard Business School, KEL059 (2006) 1-11

The reintroduction of Thalidomide brings together ethical and business issues to complicate the regulatory affairs decisions.

OTHER ARTICLES:

Higgins, Robert F., Richard G. Hamermesh and Virginia A. Fuller. "Note on the US Food and Drug Administration," Harvard Business School No. 9-807-050, Revised January 25, 2007, pp. 1-23.

Teisberg, Elizabeth Olmsted, and James Leonard, "Note on the FDA Review Process for Medical Devices," Harvard Business School No. 9-796-063, February 20, 1996, pp. 1-2.